# Identity Ladder: Population References for Inpainting and Conditional Generation

Anonymous CVPR submission

## Abstract

*Inpainting provides a controlled setting for evaluating conditional generation: visible context can be paired with withheld content, population references, and explicit changes to individual detail. Identity Ladder organizes these references into a metric audit using brain MRI, fundus photographs, faces, and iris images. Comparisons include repeat captures, bilateral anatomy, relatives where available, context-selected donors, independent means, attenuation, and incorrect-detail controls. They separate reference distortion, residual magnitude, target alignment, and acquisition sensitivity without imposing one biological ordering. In the provisional brain panel, mean squared error places the dataset mean at 0.413 and the mirror at 0.717 on a repeat-zero, stranger-one scale. In CelebA, a selected donor scores 0.399 under local pixel error but 0.983 under a whole-image identity reference on self-zero scales. The comparison changes spatial scope and is not a generated-composite identity result. Masked Gaussian maps localize differences, and validity masks retain missing support as missing. We propose applying the same controls to generated completions within prespecified target populations, reporting raw, common-reference, and matched-reference profiles with coverage and uncertainty. Target strata, donor cohorts, and acquisition domains remain distinct. The existing experiments characterize metrics and references; population-stratified generator behavior is not yet measured, and brain/fundus evidence remains provisional. The framework supplies testable controls for inpainting and a defined route to broader conditional-generation audits where an identity-bearing condition and meaningful reference unit are available.*

## 1. Introduction

Inpainting is a conditional-generation task with a useful experimental structure: visible context is fixed while a missing region must be completed. When a real region is deliberately hidden, an observed target supports controlled comparisons. A completion can resemble the target, a population-typical image, or plausible but unrelated anatomy for different reasons. Its score therefore needs references that distinguish retained individual information from common structure and acquisition variation.

Human anatomy supplies such controls without requiring a generated image to be declared the correct answer. Repeated captures hold the anatomical unit approximately fixed while changing acquisition. Bilateral comparisons retain a person but change the anatomical structure. Related individuals may share some anatomy, while unrelated donors provide a population reference. An average of independent donors supplies a different control: it removes much individual variation while retaining common structure. These comparisons define separate experimental questions. They do not imply that every metric should place every bilateral structure or relative at one universal intermediate rung.

Identity Ladder turns these comparisons into a metric audit and a set of references for evaluating completions. Each candidate is evaluated on valid acquired support and against a denominator estimated without the target's person or family. Context selection is paired with the same selection on unrelated anatomy, and selection using the hidden region is labeled an oracle. An attenuated correct donor, an amplitude-matched incorrect residual, and white noise distinguish loss of detail from wrong detail. Reporting these conditions together makes several otherwise conflated interpretations testable.

Applied to generated outputs, the same design could characterize behavior within prespecified population slices. An audit can ask whether a model's preference for typical anatomy, attenuation, incorrect detail, or acquisition sensitivity differs between target groups. Target strata, donor reference cohorts, and acquisition domains are separate axes: an age-matched donor mean is a control, not evidence of subgroup model performance. Comparable targets, masks, references, and sampling budgets are needed before differences between groups can be interpreted.

Our contribution is an explicit comparison protocol, a source-traceable collection of donor and negative-control profiles, and a spatial reporting design across anatomical image domains. The measured experiments characterize the controls and metrics; audits of generated outputs across

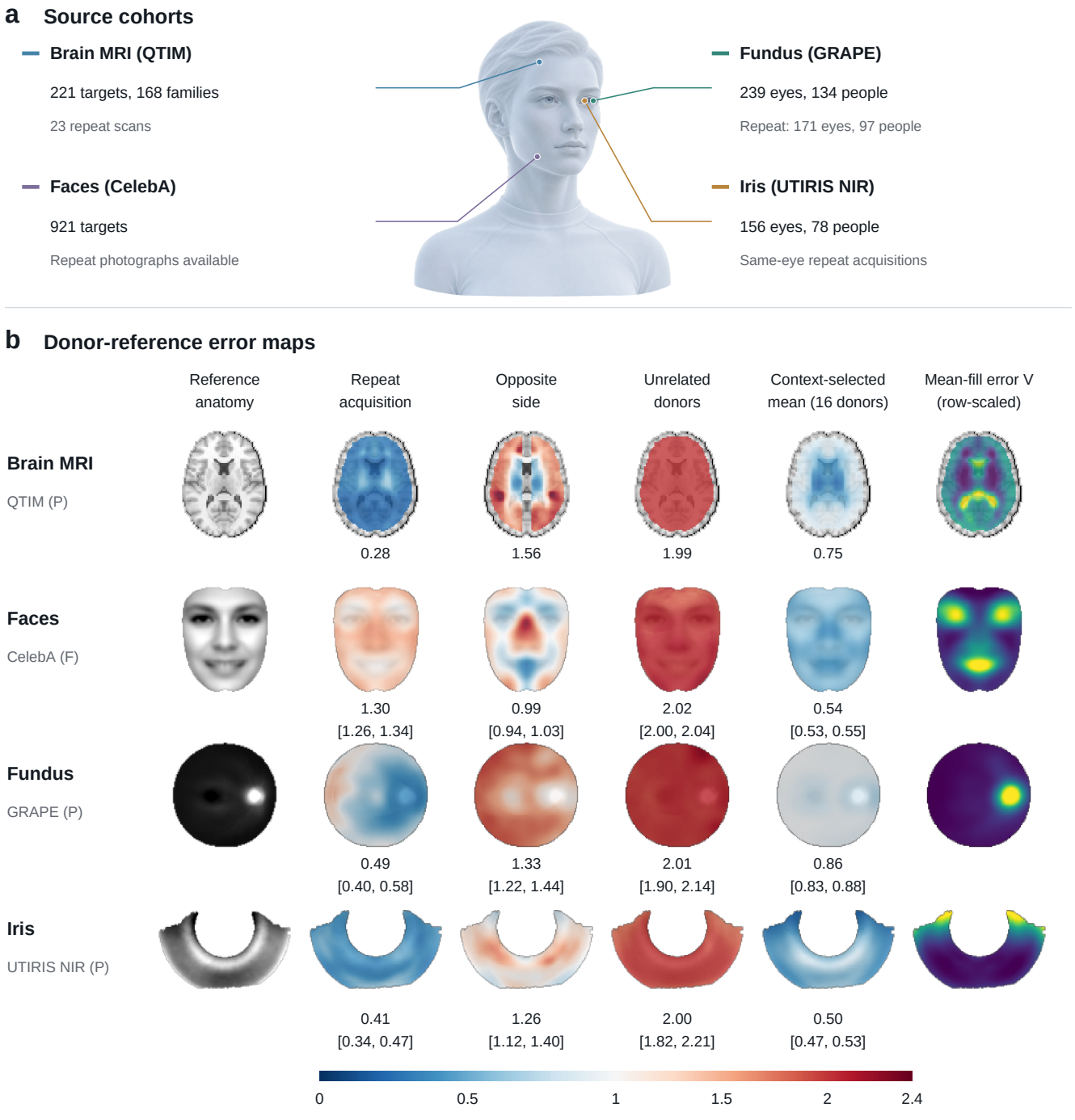



Figure 1. **Source cohorts and population-reference error maps. a,** Cohorts and counts; eligibility varies by comparison, with GRAPE repeat coverage shown separately. **b,** Observed donor controls: local squared-error ratios against an independent mean fill ($Q = 1$). References are population means or an independent MRI template. Pooled numbers differ from the anchored window-level tables; recorded intervals appear where available. Context-selected means use 16 donors; V is row-scaled. Rows use modality-specific display sizes. The pale MRI outline and shared crop are display-only; brain windows are Gaussian. Fundus uses GRAPE P8 and iris uses P2. F denotes source-frame approval; P denotes provisional evidence. The bust is decorative and synthetic; secondary datasets and display provenance remain in the supplement.

target strata remain proposed. We define that extension and map its evidence requirements without treating cross-dataset contrasts as demographic disparities. Squared error's conditional-mean optimum remains a legitimate prediction target, and uncertain hidden detail need not be uniquely recoverable. Broader conditional-generation applications require their own identity unit and valid references; unconditional generation has no intended target individual. BraTS brain inpainting [8] is one practical application, including the distinction between an observed healthy reference and unobserved counterfactual tissue.

## 2. Related work

**Reconstruction and perceptual quality.** The perception–distortion tradeoff separates reference distortion from distributional perceptual quality [1]. Learned Perceptual Image Patch Similarity (LPIPS) evaluates aspects of visual similarity that pixel distances miss [18]. Neither objective alone defines whether the retained detail belongs to the target person. Our audit adds controlled person-, organ-, and acquisition-level comparisons to this distinction. Structural similarity (SSIM) supplies a local reference comparison [16], but its interpretation still depends on the anatomical and acquisition changes represented by a pair.

**Anatomical completeness and harmonization.** CARE evaluates sparse-view CT reconstruction using anatomy-aware measurements of large and small organs, intestines, and vessels [10]. Its objective is structural completeness rather than validation of a donor-based identity scale. LMSB addresses anatomical changes in OCT harmonization by learning an anatomy-guided latent metric before distribution matching [17]. It demonstrates why Euclidean proximity in appearance need not coincide with anatomical consistency. Our contribution is an evaluation design around such interpretations, not an empirical comparison against either reconstruction or harmonization system.

**Anatomical representations and biometric units.** Brain-ID learns contrast-agnostic anatomical features from synthetic inputs and evaluates robustness and downstream brain-imaging tasks [11]. Those results motivate acquisition controls, but a Brain-ID distance has not been completed in this study. Iris recognition illustrates a complementary distinction: a person has two eyes, and person membership does not make their iris textures one biometric instance [2, 3]. The relevant identity unit must therefore be specified before an anatomical similarity result is interpreted. ArcFace supplies an identity-oriented face representation [4]; our local-donor analysis distinguishes that whole-image reference from the window-level reconstruction distance.

**Task-specific metric validation.** Metrics Reloaded emphasizes matching validation measures to the task and data [13]. We operationalize that principle through separate controls for attenuation, incorrect detail, bilateral correspondence, and acquisition. A successful check on one axis does not establish success on the others.

## 3. Methods

### 3.1. The donor ladder and the unit

Let $x_{p,t,s}$ denote an image of person $p$, acquisition $t$, and anatomical side $s$. The identity unit is a person for the face comparison and an eye for the iris comparison; bilateral structures remain separate observed anatomy in every domain. For a target region, candidates include another acquisition, the actual opposite-side structure in a fixed common frame, related donors where labeled, unrelated donors, metadata-matched means, and the dataset mean. Unavailable relationships remain unavailable.

Let $\hat{\mu}_{-g}$ be a masked population mean excluding the target's person or family $g$. For each candidate $f_i$, squared error and the reference error are evaluated on the same eligible support $E_i$. Define

$$e_i(f) = \frac{\sum_{v \in E_i} (x_i(v) - f_i(v))^2}{|E_i|}, \qquad V_i = e_i(\hat{\mu}_{-g}), \qquad Q(f) = \frac{\sum_i e_i(f)}{\sum_i V_i}. \tag{1}$$

The ratio is formed after summing errors, not by averaging unstable per-target ratios. The dataset mean has $Q = 1$ by construction on that support. Under an equal-variance additive model with an independent stranger, its expected squared error is twice the population-mean error. A finite estimated mean, unequal variances, support changes, and nuisance correlations can alter this reference. The stranger level of two is not a theorem for SSIM, LPIPS, or an arbitrary representation.

For oriented distances, we also retain the recorded panel scales

$$R_{\text{rep}}(f) = \frac{\bar{d}(f) - \bar{d}(\text{repeat})}{\bar{d}(\text{strangers}) - \bar{d}(\text{repeat})}, \qquad R_{\text{self}}(f) = \frac{\bar{d}(f) - d(x,x)}{\bar{d}(\text{strangers}) - d(x,x)}. \tag{2}$$

The brain panel uses the repeat anchor, whereas face and iris use the self anchor and retain repeat as a measured acquisition floor. These coordinates are not interchangeable biological units. Unstable or undefined denominators are not replaced by zero. MSE is a raw squared intensity distance; SSIM and correlations are higher-is-closer

Table 1. **Measured references and proposed generator audits.** Twelve audit questions are grouped here and detailed in the supplement. These are qualitative assessments of experimental scope, not completion rates or measured group performance. The table is generated from `population_audit.json`; measured evidence remains in the modality tables and implementation probes.

| Requirement | Available evidence | Extent |
|---|---|---|
| Target and population | Identity units and independent donor pools | Defined; strata audit proposed |
| Measurement contract | Masks; SSIM axes; PSNR clipping probes | Code/proxy checks |
| Lost or wrong detail | Means, attenuation, wrong residuals | Mechanism; MRI provisional |
| Acquisition sensitivity | Repeats, contrast and pupil controls | Partial/provisional |
| Structure and scale | Label toys, maps, earlier size sweeps | Proxy/partial |
| Generated group profiles | Matched donor references; no group outputs | Not measured |
| Sampling and decisions | No model-rank or seed-diversity validation | Proposed validation |

scores that the anchored coordinates orient toward increasing distance. Residual correlation subtracts the independent dataset mean before computing Pearson correlation. The TM-like and lDDT-like measures adapt protein-comparison ideas [14, 19] to intensities, rather than transferring their original structural interpretation. Their kernels and independently estimated repeat tolerances are defined in the supplement. The saved unrelated-reference rank is $c = P(d_{\text{stranger}} > d) + \frac{1}{2}P(d_{\text{stranger}} = d)$, computed against the recorded donor panel. It expresses relative discrimination, not the probability that identity was preserved.

### 3.2. Proposed population audit of completions

For a held-out target $x_i$, context $c_i$ and mask $M_i$, let $y_{ik} = G(c_i, M_i, z_k)$ be completion $k$. For target stratum $a$, lower-is-closer paired distance $d_m$ and control $f_{ir}$, a proposed profile is

$$\Delta_{a,m,r} = \mathbb{E}_{i|a}[\mathbb{E}_k d_m(y_{ik}, x_i; E_{ir}) - d_m(f_{ir}, x_i; E_{ir})] . \quad (3)$$

Support and target eligibility are fixed independently of generated content, with model failures reported separately. Seeds are averaged within targets before the population average. Negative values mean metric preference over the control, not identity preservation. Profiles retain raw distances and both common and stratum-matched references, since group-specific normalization can conceal raw differences. Fill-only, identically composited and whole-image endpoints require explicit spatial scope; set-level FID/FWD need a separate definition. Intervals must account for independent people/families, repeated masks and outputs. The current experiments characterize controls; generated subgroup profiles in Eq. 3 have not been measured.

### 3.3. Result status

The superscript $^{\text{F}}$ identifies a frozen result whose recorded frame passed the author's visual gate. It does not mean independent replication or clinical validation. The superscript $^{\text{P}}$ identifies provisional evidence with an unresolved gate or a source-version limitation. We distinguish *not run* from an undefined mathematical quantity, an unavailable relationship, and an executed result absent from the accessible snapshot. Historical outputs are not silently relabeled as newer experiments.

Every image has a validity mask. Errors, normalization, donor means, context comparisons, and local maps exclude unobserved support. Padding and stored background zeros are never measurements. A donor mean averages only valid contributing samples at each location and is undefined where no donor contributes. The acquisition field is retained where the recorded canvas permits it; version-specific crop losses remain limitations.

### 3.4. Controls, spatial maps, and uncertainty

Context-selected look-alikes use only the visible shell or ring. Oracle donors use the true hidden region and are upper-bound diagnostics rather than achievable inpainting methods. The same selection rule on unrelated anatomy controls the benefit of searching many candidates. We exclude block-search outputs that predate their missing-support correction.

Attenuation controls have the form $\hat{\mu} + \lambda(f - \hat{\mu})$, with the recorded $\lambda$ derived from the original donor comparison. Incorrect-detail controls use a stranger residual made orthogonal to the target residual and matched in norm on eligible support. Residual amplitude and target projection distinguish reduced variation from variation in the wrong direction. Residual correlation alone is invariant to positive rescaling and cannot establish preservation of amplitude.

Drawn maps use masked Gaussian local averages with standard deviation $w/\sqrt{12}$, where $w$ is the recorded window width. Box support fractions may define eligibility, but box filters do not define displayed errors. The latest geometry protocol specifies $w = fL$ with $f \in \{1/16, 1/8, 1/4\}$ and anatomical extent $L$; this specification is not evidence that every size sweep has finished. Independent family or person halves assess spatial reproducibility, and coverage is reported separately from the error map.

The source protocols use bootstrap resampling of families or people, with positions resampled independently for the panels that specify two-way intervals. Saved intervals use 2,000 resamples. Repeated regions and both eyes do not multiply the number of independent people. Fewer than ten independent units support descriptive reporting only. The recorded one-sided tests ask whether the repeat is better than the mean (T1), its attenuated version (T2), incorrect detail (T3a), and noise (T3b). The brain and iris sources retain their own Holm correction families. Failure to establish one directional test is not proof of equivalence or of its reverse.

### 3.5. Brain MRI (QTIM)

The QTIM panel contains 221 prepared targets from 168 families, including 23 repeat acquisitions. Volumes are N4-corrected and affinely aligned to an independent symmetric MNI frame, with intensity standardized within the brain mask. The panel evaluates both hemispheres at the recorded random regions and frozen deep and parietal boxes. Family members are excluded from the target's reference pool. The source records related-donor labels; it does not justify converting unknown zygosity into monozygotic or dizygotic classes.

The current tier-1 reference uses a 16 mm window and 4 mm context shell. The latest object-relative amendment gives a registered-space extent of 127.0 mm and specifies 8, 16, and 32 mm windows. The displayed fine maps and coarse look-alike maps follow the Gaussian amendments. The metric panel nevertheless remains provisional because the registration gate has not been met. The completed HACA3+ stage-1 contrast sensitivity is reported separately and does not establish a harmonized metric benchmark.

### 3.6. Fundus photographs (PAPILA, GRAPE)

GRAPE [7] supplies longitudinal and fellow-eye photographs; PAPILA [9] supplies paired eyes with expert disc contours. The protocols distinguish a GRAPE disc–fovea similarity frame from the PAPILA disc-anchored frame. The protocol versions specify green-channel preprocessing and standardization on valid field of view. Laterality comes from metadata, and opposite eyes are transformed consistently with their comparison references.

The primary frame-only aggregate was recovered from the frozen bilateral-branch Git snapshot. It uses GRAPE P8 and PAPILA P9 in a 128-pixel frame, with no pairwise registration or target-dependent intensity fit. The saved P7 snapshot is retained as historical evidence, rather than relabeled as P8/P9. P8 and P9 change the field mask and preserve more of the PAPILA acquisition field. Fixed-composition map regions hold the contributing target set constant over space. This avoids interpreting abrupt changes of cohort membership as anatomical boundaries. The saved P7 block searches predate the support correction and are excluded. No value from a currently running fundus metric panel is inferred here. The completed v1 panel is kept separate from its newer extension; its reference donor is the repeat for GRAPE and the fellow eye for PAPILA, where T1 is unavailable.

### 3.7. Face photographs (CelebA, IMDb-WIKI)

The frozen v6f panel analyzes CelebA [12] and IMDb-WIKI [15] separately. The available protocol specifies a landmark-affine population frame, image-wise masked intensity normalization, restricted pose, and fixed local windows with context rings. CelebA target and pool identities use separate dataset partitions. The IMDb-WIKI analysis clusters recorded kinship links and excludes target-family members from the pool. The base face pipeline and its complete v6f protocol were not committed in this source snapshot, which limits reconstruction of the upstream selection.

The face repeat is another photograph chosen by the recorded pose/expression rule, rather than an exact duplicate. The bilateral candidate is the opposite face half in the same photograph, with a midline exclusion. The panel's local pixel distances are distinct from its ArcFace donor reference. The latter compares whole-image embeddings of target and donor and is unavailable for means, unembedded reflections, and artificial controls. It is not an embedding of the composited inpainted region.

### 3.8. Iris images (UTIRIS, UPOL)

UTIRIS [6] near-infrared images are primary, with UTIRIS visible-light and UPOL [5] reported separately. The P2 frame uses learned iris segmentation, a normalized radial–angular strip, and masks for acquisition support, lids, and reflections. The same-eye repeat is the next eligible capture in fixed source order rather than the closest image by a metric. The fellow-eye strip is transformed by the recorded angular reflection. Donor pools exclude the target person.

The panel retains measured coverage and endpoint-specific eligibility. It uses no fixed-sector removal or pairwise rotation search. Lid shadow remains a known source of shared appearance, and the saved status records an outstanding final mask-sheet review. Earlier iris-code calibration is background evidence and is not substituted for this strip-based panel. The full P3 extension reported by the workstation handoff includes LPIPS, Betti matching, Fréchet inception distance (FID), and Fréchet wavelet distance (FWD), but is not contained in the accessible aggregate file.

### 3.9. Synthetic simulations

Saved synthetic tissue and eye experiments supply mechanism controls. Their known-target shrinkage path verifies attenuation coordinates by construction and does not

independently validate an identity metric. A threshold segmenter supports conditional observations about label overlap and simple topology; it is not a trained-segmenter result. Synthetic variance components are illustrative, not estimates of human heritability. These shared experiments remain secondary to the cross-anatomy metric audit.

# 4. Results

## 4.1. An inpainting-evaluator case study

Saved synthetic probes of BraTS `inpainting` 0.0.12 identify a two-spatial-axis SSIM convention and PSNR clipping that breaks exact rank equivalence with MSE. They motivate an explicit implementation contract, not a claim of independent anatomical sensitivity. The supplement retains both dependency-version records and all numerical probes; no generator or organizer-server validation is implied.

## 4.2. Brain MRI (QTIM)

All brain-panel findings in this section are provisional. Table 2 reports the saved random-region, both-hemisphere profile, with source two-way intervals. The MSE position of the dataset mean is below the mirror position, while the corresponding ordering changes under TM-like distance. The difference is an observed metric profile under the recorded alignment, not evidence that a population mean preserves more of a person's anatomy.

The tests also distinguish separate requirements. MSE establishes that the repeat is better than the dataset mean, incorrect detail, and noise in the recorded directional tests. It does not establish that the repeat is better than its attenuated version. The attenuated-repeat MSE position is $-0.014$ with interval $[-0.028, -0.004]^{\mathrm{P}}$, where the original repeat is the zero anchor. For the repeat-tolerance lDDT-like measure, the corresponding position is $-0.084$ with interval $[-0.108, -0.065]^{\mathrm{P}}$. These findings show a possible attenuation preference while leaving the registration gate unresolved. They do not make failure of T2 a general theorem about those metrics.

HACA3+ stage 1 provides a separate contrast sensitivity, not a completed identity endpoint. Across its 23 paired scans, mean absolute deep-region session shift changes from $0.121$ to $0.016$ standardized intensity units$^{\mathrm{P}}$. The parietal counterpart changes from $0.059$ to $0.033^{\mathrm{P}}$. The source records a pending visual verdict, and neither measurement establishes preservation of subject-specific detail.

## 4.3. Fundus photographs (PAPILA, GRAPE)

The frozen bilateral-branch aggregate supplies the P8/P9 frame-only profile in Table 3. GRAPE has 239 target-eye units from 134 people in the stranger comparison; the repeat comparison has 171 units from 97 people$^{\mathrm{P}}$. PAPILA has 488 target eyes from 244 people and no other-visit rung$^{\mathrm{P}}$. The dataset mean is closer than the fellow eye under squared error in both saved summaries. The context-selected mean of look-alikes can outperform the unconditional mean in this representation. These findings are provisional and do not establish the identity of the inferred detail.

The Gaussian maps in Figure 1 retain these preprocessing labels. High split-half correlation cannot establish anatomical validity if a shared preprocessing artifact is reproducible too. The completed v1 metric registry and the running newer extension remain separate. Superseded support-sensitive block searches supply no primary result.

## 4.4. Face photographs (CelebA, IMDb-WIKI)

Table 4 reports the available v6f tier-1 profiles on the self-anchored scale. The repeat photograph has substantial pixel error despite preserving person membership. The dataset mean is closer than the repeat under MSE in both datasets, while a local context-selected donor is closer still. That pixel preference does not identify the donor as the target person.

For CelebA, the selected donor's local MSE position is $0.399^{\mathrm{F}}$, while its whole-image ArcFace donor-reference position is $0.983^{\mathrm{F}}$. The corresponding IMDb-WIKI values are $0.397$ and $0.984^{\mathrm{F}}$. These are different measurements at different spatial scopes. Their disagreement shows that local photometric matching can select a donor whose global identity representation remains near the unrelated reference. It does not measure identity retention by a completed composite reconstruction, and no ArcFace value is assigned to an unembedded mean.

## 4.5. Iris images (UTIRIS, UPOL)

The current P2 strip panel contains 156 target eyes from 78 people for UTIRIS NIR. The secondary UTIRIS VIS and UPOL panels contain 158 eyes from 79 people and 128 eyes from 64 people, respectively$^{\mathrm{P}}$. The VIS repeat endpoint has one fewer eligible eye, which is retained as endpoint-specific coverage rather than silently changing every comparison. Table 5 reports mean-relative squared error on the support defined for each condition. The repeated eye is closer than the fellow eye in all three saved summaries, and a context-selected unrelated donor can fall between them. These are pixel-space relationships; they are not an iris biometric verification result.

The source retains distinct $R_{\mathrm{self}}$ and $R_{\mathrm{rep}}$ values, window-level controls, and the directional tests. It also records a lid-shadow mask review limitation. The newer full P3 run is absent from this aggregate snapshot. Consequently, its LPIPS, Betti-matching, FID, and FWD outcomes are not numerically reconstructed from P2 or from old iris-code results.

Table 2. **Provisional brain audit profile.** $R_{\text{rep}}$: repeat zero and mean of the recorded unrelated donors one; lower is closer. Cells contain means and source 95% intervals, with family and position resampling. Non-repeat rows involve 221 targets from 168 families; the repeat anchor involves 23 targets. Residual correlation at the exact dataset mean is undefined. Source: `results/donor_ladder_brain_metrics.json`.

| Metric | Dataset mean | Mirror | Incorrect detail | Noise |
|---|---|---|---|---|
| MSE | $0.413^{P}$ [0.367, 0.443] | $0.717^{P}$ [0.622, 0.792] | $1.006^{P}$ [0.992, 1.020] | $0.921^{P}$ [0.893, 0.944] |
| SSIM | $0.738^{P}$ [0.680, 0.784] | $0.823^{P}$ [0.755, 0.873] | $1.023^{P}$ [1.014, 1.036] | $1.132^{P}$ [1.088, 1.194] |
| Residual correlation | *undefined* | $0.752^{P}$ [0.688, 0.808] | $1.003^{P}$ [0.996, 1.009] | $1.003^{P}$ [1.000, 1.006] |
| TM-like | $0.723^{P}$ [0.642, 0.778] | $0.667^{P}$ [0.530, 0.761] | $0.997^{P}$ [0.987, 1.007] | $1.075^{P}$ [1.041, 1.121] |
| lDDT-like, repeat tolerance | $0.355^{P}$ [0.254, 0.430] | $0.890^{P}$ [0.843, 0.925] | $1.050^{P}$ [1.030, 1.080] | $2.073^{P}$ [1.924, 2.273] |

Table 3. **Provisional frame-only fundus profile.** Mean-relative squared-error ratio $Q$, with source person-bootstrap intervals; lower is closer. These P8/P9 map-compatible summaries differ from the older P7 support and from the window-level metric panel. PAPILA has no repeat observation. Source: `results/donor_ladder_2d_facefmt.json` at the frozen bilateral snapshot, copied without change into the paper inputs.

| Dataset | Repeat | Fellow eye | Context mean | Stranger |
|---|---|---|---|---|
| GRAPE P8 | $0.489^{P}$ [0.404, 0.585] | $1.330^{P}$ [1.224, 1.437] | $0.856^{P}$ [0.832, 0.880] | $2.012^{P}$ [1.897, 2.140] |
| PAPILA P9 | *unavailable* | $1.467^{P}$ [1.422, 1.512] | $0.935^{P}$ [0.914, 0.955] | $1.992^{P}$ [1.941, 2.049] |

### 4.6. Synthetic simulations

The saved synthetic experiments isolate a complementary ambiguity. On the known-target attenuation path, an image can keep a high unrelated-reference rank while its residual amplitude falls. An amplitude-matched incorrect residual shows why distance from the mean alone is insufficient. Synthetic pupil dilation holds latent eye parameters fixed while changing appearance, exposing the opposite failure direction. These controls explain mechanisms, but they do not establish how often the same effect occurs in measured anatomy. The synthetic label experiments additionally depend on the anatomical prior and segmentation construction. Their conditional findings do not establish a universal Dice or topology result.

## 5. From reference audits to model audits

The existing experiments characterize reference candidates and metric behavior. Table 1 separates that evidence from the proposed audit of conditional generation across populations. Coverage describes what has been investigated, not a percentage of a model-evaluation problem solved.

**What the references establish.** Means, attenuation, incorrect residuals, and repeats expose different interpretations of a similarity score. Symmetry supplies a comparison condition rather than an identity label: a face reflection shares capture conditions, a fellow iris changes the biometric instance, and hemispheres require correspondence. Cross-anatomy observations stress these interpretations without establishing a universal ordering or transfer to every inpainting task.

**How model behavior could be tested.** For a fixed context and mask, generated outputs can be compared with the same ladder controls using Eq. 3. Repeating this comparison within target strata could reveal whether a model's scores approach global or matched means, retain target-aligned variation, or respond to acquisition differently between groups. The context must support the claimed detail: a conditional mean can be a legitimate prediction when the withheld content is ambiguous. Residual correlation misses positive amplitude shrinkage, and stranger calibration supplies no new anatomical information. The profile therefore retains complementary controls rather than identifying one coordinate with preserved identity.

**Population and sampling interpretation.** Global and matched references answer different questions; group-specific normalization alone can conceal raw gaps or dif-

Table 4. **Face metric and donor-identity profiles.** Recorded $R_{\text{self}}$ point estimates; target self zero, mean of unrelated donors one. The source retains family/position bootstrap intervals; full records accompany the preparation manifest. The approved frame has 921 CelebA targets and 678 IMDb-WIKI targets, respectively clustered into 921 and 453 units. ArcFace compares whole target and donor images; the pixel metrics compare local windows. Source: `results/face_ladder_metrics.json`.

| Dataset | Metric | Retest | Bilateral | Dataset mean | Context donor |
|---|---|---|---|---|---|
| CelebA | MSE | $0.640^{F}$ | $0.475^{F}$ | $0.499^{F}$ | $0.399^{F}$ |
| CelebA | SSIM | $0.828^{F}$ | $0.693^{F}$ | $0.844^{F}$ | $0.726^{F}$ |
| CelebA | ArcFace donor reference | $0.354^{F}$ | *unavailable* | *unavailable* | $0.983^{F}$ |
| IMDb-WIKI | MSE | $0.690^{F}$ | $0.529^{F}$ | $0.498^{F}$ | $0.397^{F}$ |
| IMDb-WIKI | SSIM | $0.856^{F}$ | $0.739^{F}$ | $0.859^{F}$ | $0.743^{F}$ |
| IMDb-WIKI | ArcFace donor reference | $0.396^{F}$ | *unavailable* | *unavailable* | $0.984^{F}$ |

Table 5. **Provisional P2 iris donor ladder.** Squared-error ratio $Q$ against an independent dataset mean evaluated on the same support; lower is closer. The mean anchor is one by construction; an independent equal-variance stranger has theoretical expectation near two. Intervals use person bootstrap. Target-eye/person counts are 156/78, 158/79, and 128/64; the VIS repeat endpoint uses 157 eyes. Source: `results/iris_ladder_metrics.json`, results block.

| Dataset | Same-eye repeat | Fellow eye | Context donor | Stranger |
|---|---|---|---|---|
| UTIRIS NIR | $0.407^{P}$ [0.345, 0.475] | $1.256^{P}$ [1.123, 1.395] | $0.599^{P}$ [0.540, 0.660] | $2.001^{P}$ [1.816, 2.207] |
| UTIRIS VIS | $0.569^{P}$ [0.503, 0.640] | $1.293^{P}$ [1.211, 1.375] | $0.861^{P}$ [0.826, 0.898] | $2.018^{P}$ [1.884, 2.176] |
| UPOL | $0.177^{P}$ [0.154, 0.200] | $1.118^{P}$ [1.037, 1.198] | $0.841^{P}$ [0.798, 0.882] | $2.138^{P}$ [2.007, 2.281] |

ferences in reference variance. A group contrast characterizes the model–metric system; attribution requires checking reference, acquisition and task-composition effects. Mask difficulty, location, acquisition, donor-pool size and missing outputs must remain visible in a comparison. Repeated outputs distinguish variation across targets from variation across seeds for one context; reduced variance alone does not establish mode collapse. Group gaps require independent-unit uncertainty and an explicit target population, and do not by themselves establish causal discrimination or fairness. Actual generator outputs, an independent anatomy or task criterion, and held-out validation are needed to test these proposed interpretations.

**Limitations and evidence coverage.** Registration and acquisition remain central confounders in the brain and fundus panels. Source protocols have advanced beyond some accessible outputs, and preprocessing versions must not be merged by name. The evidence freeze predates receipt of the base face pipeline and full iris P3 aggregates; integration remains pending. Lid-shadow information, changing support and selection on usable windows can affect iris comparisons. The current residual and rank coordinates have not been validated as a fused identity score. Results at one window size do not establish stability over the latest object-relative size grid.

**Ethics and data.** The manuscript preparation uses existing aggregate outputs and does not read or redistribute participant images or identifiers. The anatomical map illustrations are population summaries or templates, not individual exemplars. Iris images and private row tables remain outside the public repository. The preparation record does not establish the original dataset consent or the institutional basis for each secondary analysis. Those dataset-specific approval, exemption and consent statements require author confirmation before a submission-ready ethics declaration can be made.

## 6. Conclusion

Identity Ladder supplies population references for interpreting similarity in inpainting and a defined extension to audits of conditional generation. The measured controls probe responses to detail loss, incorrect detail and acquisition variation; generated-output profiles across target strata remain to be tested. Broader applications require an appropriate identity unit, conditioning contract and reference population, preserving the distinction between a proposed audit and demonstrated model behavior.

## LLM Usage Statement

OpenAI Codex assisted with source inspection, manuscript structure and prose, bibliography metadata retrieval, aggregate-to-table preparation code, and build verification. OpenAI image generation produced the decorative anonymous bust in the provenance illustration; labels, cohort counts and measured heatmaps were assembled separately from recorded sources. The project also records Claude assistance with experimental code, protocols, interpretation and earlier writing. This manuscript-assembly session did not create experimental observations or generate participant images. The authors remain responsible for checking the scientific claims, evidence status, citations and final text.

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